FINM3013 Introduction to Financial Derivatives

Chapter 1 Exercises

Solution

1.

a) The trader sells for 50 cents per pound something that is worth 48.20 cents per pound. Gain = ($0.5000 − $0.4820) × 50,000 = $900.

b) The trader sells for 50 cents per pound something that is worth 51.30 cents per pound. Loss = ($0.5130 − $0.5000) × 50,000 = $650.

2. Yes, the trader will exercise if the asset price is greater than $30, but will cover the cost of the call option only if the price is greater than $33. The trader exercises and loses money if the price is between $30 and $33.

3.

The company could enter into a forward contract obligating it to buy 3 million euros in three months for a fixed price (the forward price). The forward price will be close to but not exactly the same as the current spot price of 1.4500. An alternative would be to buy a call option giving the company the right but not the obligation to buy 3 million euros for a a particular exchange rate (the strike price) in three months. The use of a forward contract locks in, at no cost, the exchange rate that will apply in three months. The use of a call option provides, at a cost, insurance against the exchange rate being higher than the strike price.

4.

The arbitrageur should borrow money to buy a certain number of ounces of gold today and short forward contracts on the same number of ounces of gold for delivery in one year. This means that gold is purchased for $1800 per ounce and sold for $2000 per ounce. The cost of the borrowing is $90 per ounce. A riskless profit of $110 per ounce is generated.

5.

The investment in call options entails higher risks but can lead to higher returns. If the stock price stays at $94, an investor who buys call options loses $9,400 whereas an investor who buys shares neither gains nor loses anything. If the stock price rises to $120, the investor who buys call options gains

$$2000 \times (120 - 95) - 9400 = \$40,600$$

An investor who buys shares gains

$$100 \times (120 - 94) = \$2,600$$

The strategies are equally profitable if the stock price rises to a level, S, where

$$100 \times (S - 94) = 2000(S - 95) - 9400$$

or

$$S = 100$$

The option strategy is therefore more profitable if the stock price rises above \$100.